
Ayush Gupta

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SUMMARY

Honors Computer Science student with experience in deep learning, compiler optimization, and AI systems research. Focused on applying machine learning and signal processing to efficient, real-world applications in healthcare and large-scale inference systems.

EDUCATION

B.S., Computer Science	May 2028
Barrett, The Honors College, Arizona State University, Tempe, AZ	3.83/4 GPA

TECHNICAL SKILLS

Programming: Python, Java, SQL

Developer Tools: TensorFlow, PyTorch, MLIR, Scikit-learn, Git

Certifications: Python for Data Science and Machine Learning by Jose Portilla, Deep Learning A-Z: Neural Networks and AI by Kirill Eremenko, Harvard CS 50 Intro to Computer Science

WORK EXPERIENCE

Research Volunteer MPS Lab, ASU	Oct 2025 – Present
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- Optimizing autotuners for large language model inferencing through MLIR-based compiler techniques.
- Improving model execution performance and scalability across diverse hardware architectures.

Community Assistant ASU Housing	Oct 2025 – Present
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- Working with first year students to ensure a safe and responsible environment.
- Intentionally forging connections that build an engaging residential experience.

Director of Inclusion and Leadership	Aug 2025 – Present
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- Honors College Council at Barrett (HCCB), Arizona State University
- Elected to spearhead diversity, equity, and inclusion initiatives within Barrett, The Honors College.

Research Intern Neuro Computing Lab, IIT Delhi	May 2025 – Aug 2025
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- Designed deep learning models (CNNs, ViTs) for multi-class EMG signal classification.
- Developed predictive model for hand gesture classification and arm translation
- Enhanced neural interface reliability using advanced signal processing and ML pipelines.

PROJECTS

Crowd Safety Monitoring System ASU & Scottsdale Fire Department	Aug 2025 – Present
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- Developing a real-time computer vision system to detect potential crowd crush scenarios by analyzing wave and density patterns in live video feeds.
- Implementing motion tracking and anomaly detection algorithms to provide early alerts for emergency response teams.

Autonomous Car with Sensor-Guided Navigation ASU	Oct 2024 – Dec 2024
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- Built a car with touch sensors, a color sensor, and a gyroscope, integrated with MATLAB.
- Developed a MATLAB program for autonomous maze navigation and path tracking.